*see also* Great Terror: Nuclear weapons research
Space:
foamlike structure in quantum gravity, 477–78, 478f, 494–97, 495f
contraction of: *see* Length contraction
*see also* Absoluteness of space and time; Relativity of space and time; Space warpage
Space warpage (curvature):
concept deduced by Einstein, 107
concept explained, 108–9, 108f
as due to rubbery rulers, 397–400, 399f
*see also* Black-hole properties; Embedding diagrams; Gravitational waves; Wormholes
Spacetime:
concept formulated by Minkowski, 87–88, 92–93, 414
absoluteness of, 90, 92, 91b-92b, 414
space and time derived from, 90, 92
Spacetime curvature (warpage):
concept deduced by Einstein, 107–8
equivalent to tidal gravity, 110–12, 110f, 112f
produced by mass, energy, and pressure, 113–19, 118b-19b
as due to rubbery clocks and rulers, 397–400
objects made from, overview of, 523–24
Spacetime diagrams:
in flat spacetime, 74–76, 75f, 77b, 91b
in curved spacetime, 249–51, 251f
for a star that implodes to form a black hole, 249–51, 251f, 281, 282f, 415b
for the growth of a black hole when matter falls in, 418b
for the fall of an astronaut into a black hole, 452f
for the expanding universe, 460–61
Special relativity:
Einstein's formulation of, 80–83
experimental tests of, 78, 83–84, 566N78
Spectrum of electromagnetic waves, 25f, 379
Speed of light:
as maximum speed anything can travel, 82
*see also* Absoluteness of speed of light
Stability and instability:
of cold, dead stars, 203–5, 204b
of black holes, 296–98
of formation of singularity by imploding star, 453–56, 455f, 459, 466, 473–74
of travel through black hole to another universe, 473–74
Star:
laws governing structure of, 143–44, 144f
squeeze/pressure balance in, 135f, 137, 143–45, 144f, 151, 154f, 156b-57b, 160, 161f, 175–77, 190, 199, 202, 253
origin of heat, 182, 184, 191; *see also* Fusion, nuclear
cold, dead stars, 198, 203f, 202–6, 203n, 204b
unstable stars, 203–5, 204b
supermassive stars, 341
*see also* Black holes; Neutron stars; Sun; White dwarf stars
Stroboscopic sensor, 376
Strong force, *see* Nuclear force Sun: